

AIPV 2026 slides (scaffold)

Inverting the formalization workflow — replace with final title

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May 2026

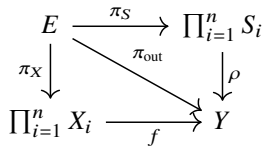
Outline

- 1 MPC at a glance
- 2 How this project started
- 3 Method
- 4 Glossary (non-Rocq audience)

Multi-party computation (high level)

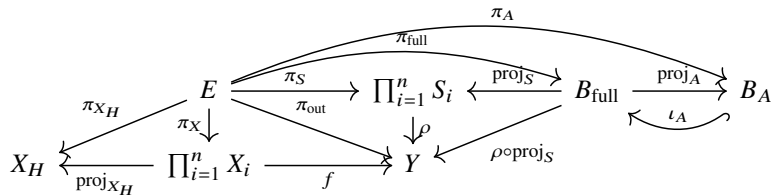
- Parties hold **private inputs** and want a function of those inputs.
- Security: limit what an adversary learns from their **view** (messages, corrupt parties).
- **Correctness:** local pieces + reconstruction $\Rightarrow$ same result as ideal $f(\text{inputs})$.

MPC correctness diagram



Correctness: $\rho \circ \pi_S = f \circ \pi_X = \pi_{\text{out}}$.

MPC ambient diagram (adversarial views)



Background (before covering spaces)

- I had **not** formally studied topology, group theory, or covering spaces.
- I did know the **MPC-shaped goal**: several parties, local data, global reconstruction, control what leaks.
- I started with a **rough search**: what mathematical objects “look like” that protocol pattern?

Covering spaces and a surjective hook

- **Covering spaces** stood out: a **covering map** is a surjection with structured fibers/sheets—reconstruction and hidden structure are already part of the picture.
- That surjective map was what **caught my attention** first (not a claim of prior domain mastery).

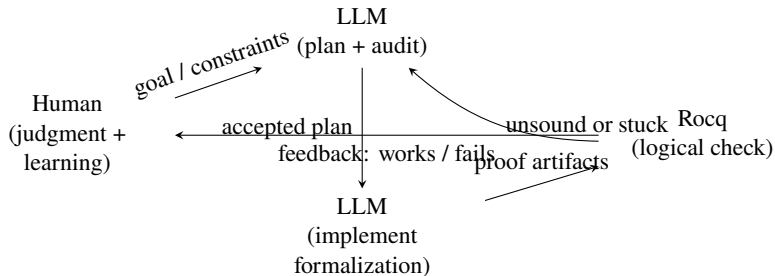
Hypothesis and workflow choice

- **Working hypothesis:** any **surjection** (more abstractly, an **epimorphism** in a suitable category) between structured sets might admit a “secret hiding + authorized reconstruction” reading worth testing.
- Rather than spending years on pen-and-paper prerequisites first, I began to **develop and practice formalization as a prototyping method**: state the connection in Rocq early, let the checker compress nonsense quickly, and **learn the mathematics on demand** from each iteration.

LLM workflow (implementation loop)

- ① **Plan drafting:** propose a formalization plan for the target math idea.
 - ② **Plan audit:** analyze why the plan should work, where it may fail, and what parts are hard; use related-work search + reasoning.
 - ③ **Human gate:** if the human accepts the audited plan, start formalization implementation in Rocq.
 - ④ **Failure recovery:** if the strategy is found logically unsound or stalls during proofs, propose a revised plan and re-audit.
- **Human role:** judgment *and* learning from each iteration.

Roles and plan-replan feedback loop



- **Replan trigger:** failed obligations can imply missing assumptions, wrong statements, or weak automation; the LLM re-audits and proposes a new plan.

Proof assistant Environment for writing checked mathematics (here: Rocq).

Machine-checked proof A proof the assistant has fully verified.

Axiom Assumption accepted without proof (stated explicitly).

Placeholder proof Informal name for skipping a proof (Rocq: `Admitted`).